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Technical report
on
Functional Safety

ANSYS Tools

ANSYS RedHawk™, ANSYS RedHawk-SC™, ANSYS Totem™, ANSYS PowerArtist™, ANSYS PathFinder™

Manufacturer

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Revision history

Revision	Status	Date	Author	Modification / Description
1.0	replaced	2019-07-16	Hermann Mayer	Initial
1.1	replaced	2019-10-18	Hermann Mayer	Further tools / changed product versions
1.2	active	2019-12-17	Hermann Mayer	TCL1 correction in summary

Table 1: Revision history

List of abbreviations

Abbreviation	Description
FSM	Functional Safety Management
TCL	Tool Confidence Level
TD	Tool error detection
TI	Tool impact

Table 2: Abbreviations

1 Target of Evaluation (ToE)

In March 2017 ANSYS, Inc. requested TÜV SÜD Rail GmbH to test and certify the ANSYS Tools according to the standard listed in clause 3 of this report. The project number related to this technical report is 717514544.

The ToE is a collection of semiconductor tools suitable to support development of elements and systems up to ASIL D. See Figure 1 to Figure 3 for an overview of the application areas of the corresponding tools.

In August 2019 ANSYS, Inc. requested TÜV SÜD Rail GmbH to test and certify further tools. The project number related to this technical report is 717519699.

2 Scope of Testing

2.1 Test specimen

The subject of this report is the consideration of certain tool classification documents and manuals for ANSYS Tools, referring to the tools listed below.

- RedHawk
- Totem
- PowerArtist

With report version V1.1 further tools have been added:

- RedHawk-SC
- PathFinder

Additionally the considered product versions have changed as listed in Table 3.

The intention of the tool classification documents is to provide an analysis of potential tool errors and a rational of how these errors can be detected. The preconception of ANSYS, Inc. is that any tool might introduce faults into the tool outputs and thus potentially into the elements and systems to be developed, or may fail to reveal them. Therefore assumption is that every tool has TI2 in general. The potential faults of each tool are analyzed in a tool FMEA which is refined in the corresponding analysis document. For each tool at least one measure to prevent or detect faults in the tool output is given and a justification for the determination of TD1 is derived. The task for TÜV SÜD is to analyze whether it is reasonable to predetermine TCL1 for all tools considering the suggested measures. Furthermore, the provided documentation shall be confirmed to be compliant with ISO 26262-8 clause 11.

The ANSYS Tools are used for design and verification of semiconductor devices (power analysis/modelling, voltage drop analysis, ESD, etc.)

ANSYS's RedHawk is a full-chip cell-based power/ground design and verification product, addressing power integrity with static and dynamic analyses, power net electromigration analysis, signal net electro-migration and electrostatic discharge, from early in the design flow through verification and sign-off as shown in the Figure 1.

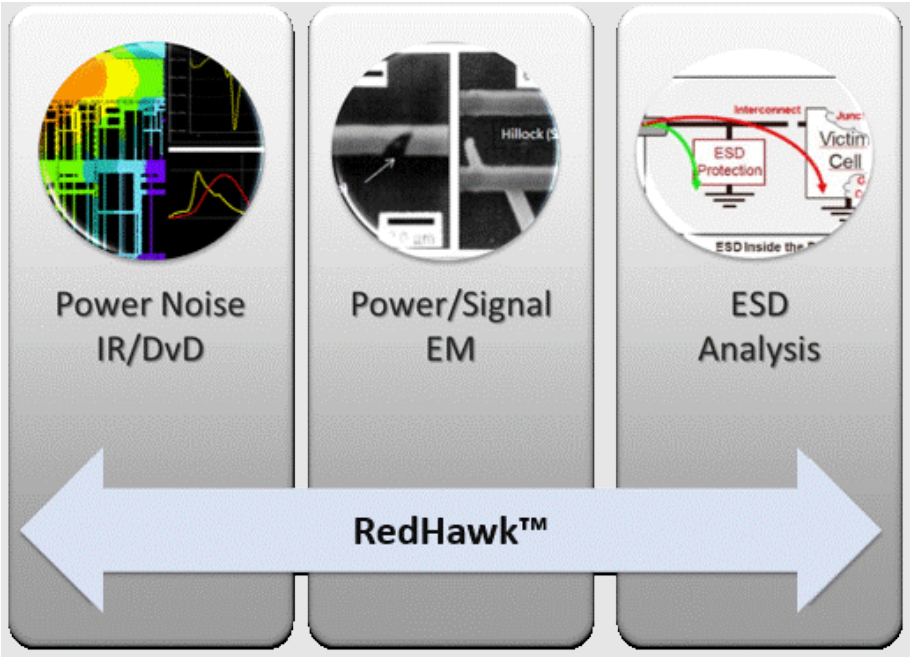

Figure 1: Overview of RedHawk flows

ANSYS's RedHawk-SC is a full-chip cell-based power/ground design and verification product, built on SeaScape platform. The flows are identical to ANSYS's RedHawk as shown in Figure 1.

ANSYS's Totem is a transistor-level product for analyzing the power integrity with static and dynamic voltage drop analysis and reliability analysis with power, signal net electro-migration and electrostatic discharge, as shown in the Figure 2.

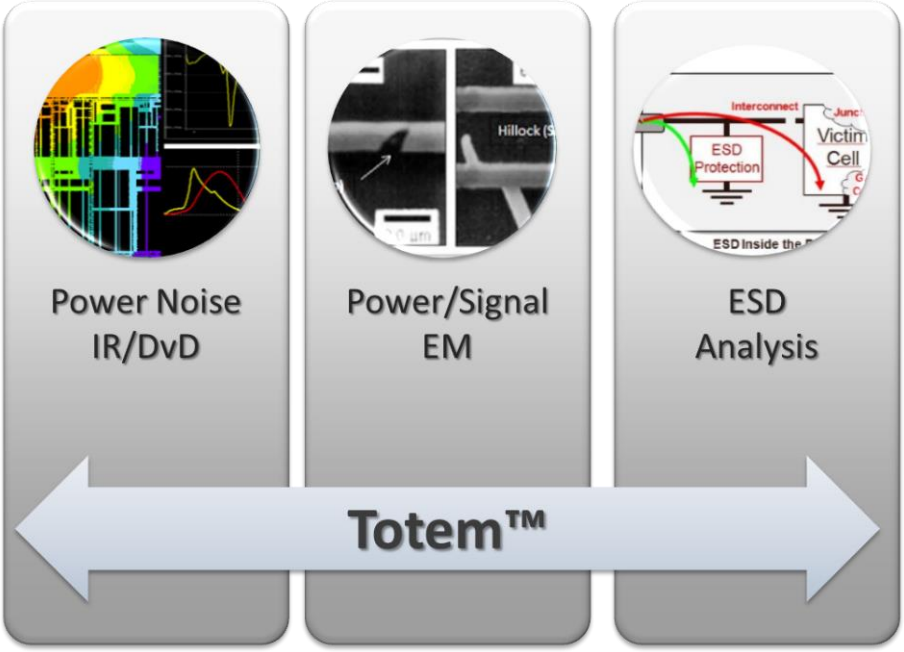

Figure 2: Overview of Totem Analysis options

ANSYS PowerArtist is a complete RTL Design-for-Power platform with fully integrated advanced power analysis, power debug, and automatic power reduction. RTL power data from PowerArtist also enables both early power grid integrity analysis during implementation and what-if analysis during system-level design, as shown in the Figure 3.

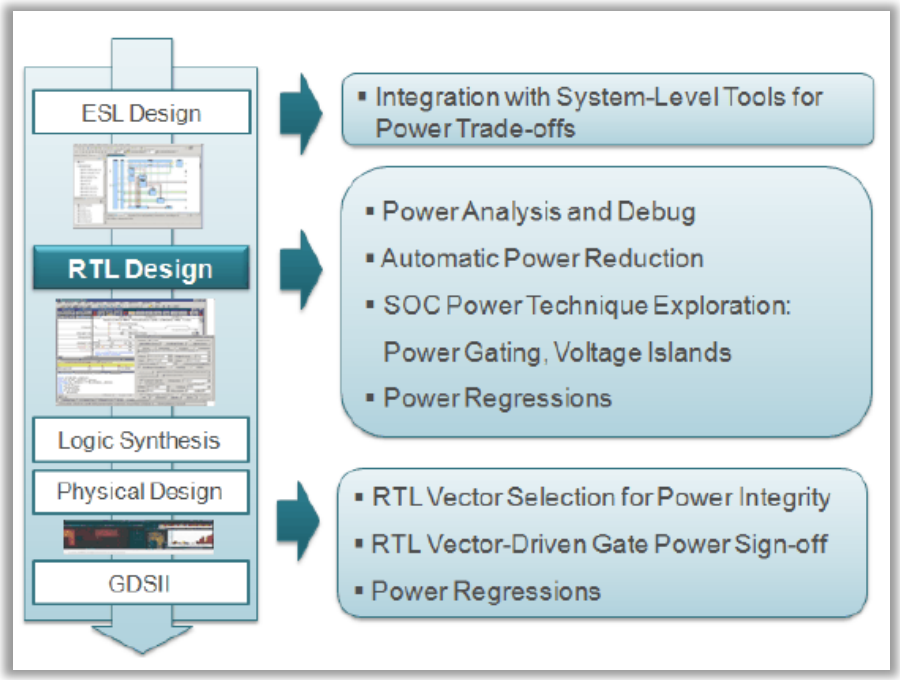

Figure 3: PowerArtist in the design flow

ANSYS's PathFinder is a ESD integrity analysis tool which is an option to RedHawk and Totem platforms, as shown in Figure 4.

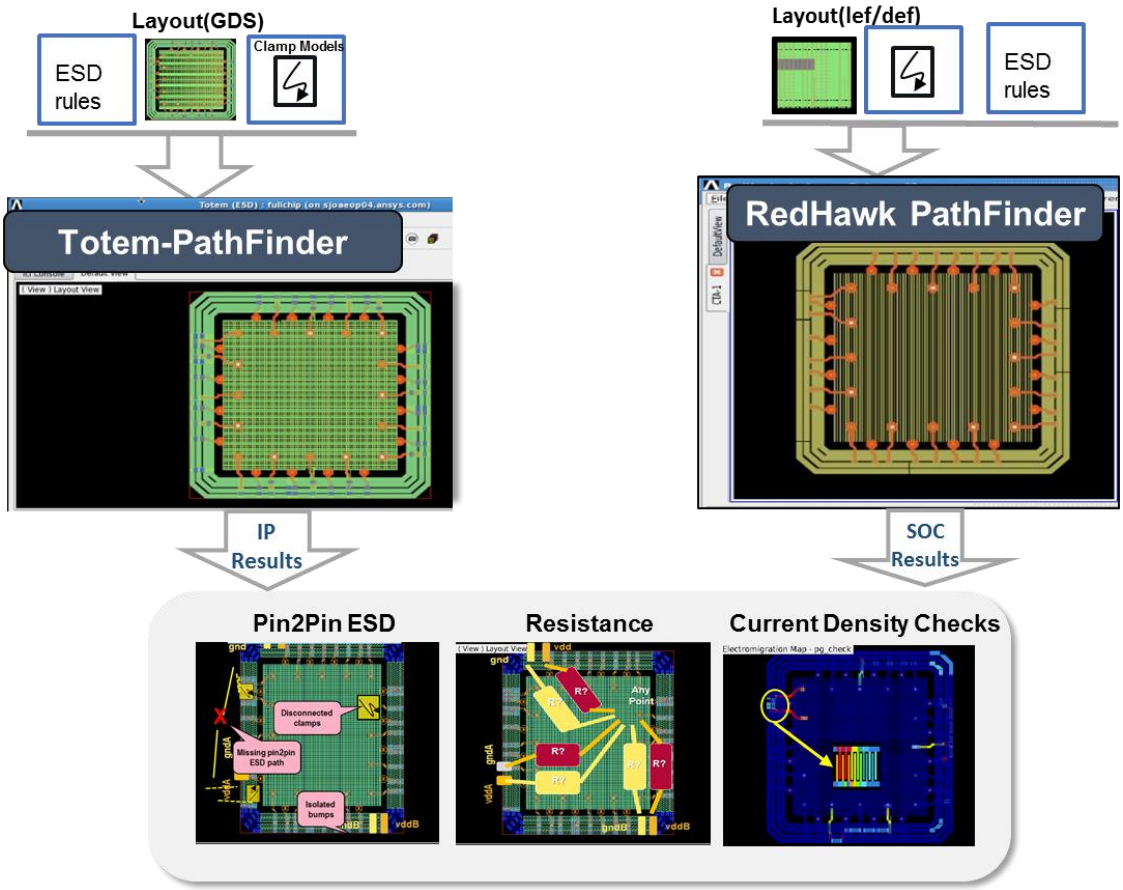

Figure 4: PathFinder ESD Analysis in RedHawk and Totem

2.2 Nomenclature of ANSYS Tools

The predetermination of the Tool Confidence Level has been performed for the tools listed in Table 3. This evaluation has taken place in the light of the tool modules as presented in the safety documentation. The report remains valid for upcoming tool versions as long as the changes do not impact the general statement of this report.

Name	Version
ANSYS RedHawk	19.1
ANSYS Redhawk-SC	2019 R1
ANSYS Totem	19.1
ANSYS PowerArtist	19.1
ANSYS PathFinder	19.1

Table 3: Identification of ANSYS Tools

3 Standards and guidelines

The regulations and guidelines which form the basis of the type testing are listed below.

3.1 Functional safety

No.	Reference	Description
/N1/	ISO 26262-8:2018	Road vehicles — Functional safety — Part 8: Supporting processes

Table 4: Functional safety

4 Documents provided for testing of ANSYS Tools

No.	Title	Document number / ID	Rev.	Date
/D1/	ANSYS RedHawk Power Integrity and Reliability Analysis	ANSYS_RedHawk_Flow_Document_for_ISO26262_TCL1_20191017.docx	1.6	2019-10-17
/D2/	ANSYS RedHawk-SC™ Power Integrity and Reliability Analysis	ANSYS_RedHawk-SC_Flow_Document_for_ISO26262_TCL1_20191008.docx	1.2	2019-10-08
/D3/	ANSYS Totem Power Integrity and Reliability Analysis	ANSYS_Totem_Flow_Document_for_ISO26262_TCL1_20191017.docx	1.4	2019-10-17
/D4/	ANSYS PowerArtist RTL Power Analysis and Reduction	ANSYS_PowerArtist_Flow_Document_for_ISO26262_TCL1_20191008.docx	1.2	2019-10-08
/D5/	ANSYS PathFinder ESD Integrity Analysis	ANSYS_PathFinder_Flow_Document_for_ISO26262_TCL1_20191017.docx	1.3	2019-10-17

Table 5: Documentation of ANSYS Tools

5 Performance and result of tests

5.1 Test reports

The following test reports were issued by TÜV SÜD Rail GmbH or other accredited test laboratories.

No.	Title	Document number / ID	Rev.	Date
[R1]	Review protocol Further Tools	RevProt FurtherTools V1.2.docx	1.2	2019-10-16
[R2]	Audit checklist for software tools according to ISO 26262	Audit checklist for software tools according to ISO 26262 ANSYS.docx	1.1	2019-10-16

Table 6: Test reports

6 Evaluation results

This clause summarizes the evaluation results with respect to the evaluation criteria.

6.1 Results according to ISO 26262-8:2018, clause 11: Confidence in the use of software tools

Each of the analysis documents listed in clause 4 (/D1/ to /D4/) has the goal to enable a user to achieve functional safety compliance satisfying ISO26262 of TCL1 by using the corresponding tool of ANSYS, Inc. The documents consist of use cases and error scenarios which may happen during analysis and measures for detecting the tool errors / malfunctions through FMEA (Failure Models and Effects Analysis). In all cases it is recommended that the user reads respective manuals of the product (user guide, reference manual, development guide) for additional information beyond this safety manual. All the analysis documents are using the same approach of description.

Inputs, outputs, workflows and functionalities are described per tool and more precisely also per use case. The tool classification analysis gives an overview of the interaction of use cases, potential malfunctions or erroneous reporting and error detection and prevention methods to achieve a TD1, and consequently TCL1.

Development, verification, validation and release of the product follow a defined development process. With each release the impact of changes to the safety documentation is internally assessed by ANSYS, Inc.

Judgment: The normative requirements are fulfilled.

Constraints: According to ISO 26262-8, 11.4.2 any predetermined tool confidence level shall be confirmed in accordance with ISO 26262-2:2018, Table 1, prior to the software tool being used for the development of a particular safety-related item or element.

6.2 Detailed Evaluation of the Assessment Base

No.	Question	Status
Q01	Is the identification and version number of the tool clearly indicated? [ISO 26262-8 11.4.4.1-a)]	OK
Q02	Are the configurations viable for safe development detailed? [ISO 26262-8 11.4.4.1-b)]	OK
Q03	Are constraints for the use in safe development detailed? [ISO 26262-8 11.4.5.1-c)]	OK
Q04	Is the purpose of the tool described? [ISO 26262-8 11.4.5.1-a)]	OK
Q05	Are safety relevant use cases described? [ISO 26262-8 11.4.4.1-c)]	OK
Q06	Are the inputs and outputs of the tool described? [ISO 26262-8 11.4.5.1-b)]	OK
Q07	Are the requirements on the tool environment detailed? [ISO 26262-8 11.4.4.1-d), 11.4.4.2-c)]	OK
Q08	Are compatibilities and incompatibilities to other tools and/or versions described? [ISO 26262-8 11.4.4.1-d), 11.4.4.2-c)]	OK
Q09	Are the features, functions and technical properties of the tool specified or a reference made to a document containing them? [ISO 26262-8 11.4.4.2-a)]	OK
Q10	Are User manual and guides available? [ISO 26262-8 11.4.4.2-b)]	OK
Q11	Are safety issues adequately highlighted in the user manual, if not contained in the safety manual? [In line with ISO 26262-8 10.2, ISO 26262-2 6.4.3.4-b)]	OK
Q12	Is the behaviour under anomalous operating conditions considered? [ISO 26262-8 11.4.4.1-d)]	OK
Q13	Are known software tool malfunctions and the appropriate safeguards, avoidance or workaround measures described? [ISO 26262-8 11.4.4.2-e)]	OK
Q14	Are the measures for the detection of malfunctions and the corresponding erroneous output of the software tool identified? [ISO 26262-8 11.4.4.2-f)]	OK
Q15	Is the safety manual precise and concise, structured in a clear manner, easy to understand by the intended users, and maintainable? [ISO 26262-8 10.4.3]	OK
Q16	Does the safety manual contain an appropriate title, an author and an approver, a unique identification of its versions, a change history, and a status? [ISO 26262-8 10.4.5]	OK
Q17	Does the safety manual contain information for communicating and receiving up-todate safety relevant information on the tool? [In line with ISO 26262-2 7]	OK

7 Summary

The test results of clauses 5 and 6 showed that the ANSYS Tools as identified in chapter 2 fulfil the requirements applicable to software development tools as given by the standards listed in chapter 3, with respect to the conditions of use of clause 4. These documents provide instructions and conditions of use to enable the classification as TCL1.

Based on these considerations, the ANSYS Tools are suitable to be used in safety-related development projects according to ISO 26262:2018 for any ASIL.

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